Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/25/2016
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Project Title:	High-Protein And Polyphenol Bar Formulations: Utilizing Whey Protein-Polyphenol Ingredients		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-0001	**Congressional District:**	4
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lila, Mary Ann Ph.D. (North Carolina State University)		
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Task Description:	The proposed research is a ground-based investigation to evaluate high protein-polyphenol bar formulation for prevention of inadequate nutrition and compromised health of crew members during space flight. Typically, high protein bars have a short shelf-life due to hardening of bars overtime. Therefore, we propose the use of protein-polyphenol particles, which will not participate in the bar hardening reaction. The bars will deliver whey proteins, a high quality protein source that will maintain muscle strength and function, and immunoprotective polyphenols from berries that can improve cognitive function and prevent oxidative stress. Complexation of polyphenols to proteins, will stabilize food quality and bioactivity for extended periods of space travel.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Consuming the necessary amount of protein to reach dietary needs can be challenging, particularly for individuals who are recovering from strenuous exercise, dieting, or experiencing space flight. Therefore, diverse, convenient protein-dense and calorie-rich food sources are needed to supplement nutrition. Protein bars can be formulated with fruits, nuts, grains, and vegetables to address nutritional and health requirements, but may fail to meet shelf life needs, due to a phenomenon known as bar hardening. Over time, soft, chewy bars transform into hard, brittle solids with undesirable texture. This hardening reaction is associated with various transformations of ingredients in bars. Mitigation of bar hardening with the use of protein-polyphenol ingredients is of interest to a wide range of individuals who need nutrients in a shelf stable for extended periods of time and those looking for the added health benefits of plant polyphenols.
Task Progress:	The goal of the project was to produce a protein-polysaccharide ingredient that was dense in nutrients and increase the quality of high-protein bars for use in space flight. We hypothesized that using a functionally “inert” protein ingredient would resist changes that cause the undesirable textural property known as “bar hardening,” which shortens the shelf life of protein bars. Particles were formed using a “green” chemistry approach where aggregation was driven by mixing whey protein isolate (WPI) with cranberry juice to form particles, separating the particles from the fluid phase, and spray drying. This provided an additional benefit of adding healthy plant polyphenols to a bar already containing high quality protein. We evaluated methods to produce protein-polyphenol particles designed for reducing bar hardness and explored potential mechanisms for reduced bar hardening. It was determined that particle formation based on co-aggregation of whey proteins with cranberry juice polyphenols (WP-CP particles) produced the most ideal particles. When used in a bar formation, WP-CP particles reduced the initial bar hardness and showed less hardening over time. The effect was thought to be the result of the particles not participating in reactions that produce bar hardening. A critical step in forming WP-CP particles was the production of protein-polyphenol aggregates that were insoluble in the pH range of 4 to 6. While there is no agreed upon mechanism for bar hardening, the proposed mechanisms are related to phase equilibration, phase separation, or network formation over time. All of these require a protein-water phase with the ability for proteins to move about and form new interactions. Just mixing protein and juice polyphenols under conditions where proteins remain water soluble – at amounts similar to the WP-CP particles – showed increased bar hardness. Therefore, the critical property may not be particle size, but, rather, removal of small reactive/native whey proteins. We propose that pre-aggregating proteins into a particle phase removes them from participating in bar hardening reactions that occur over time. The enzymatic digestion rate of the WP-CP particles was slower than whey protein isolate alone, a finding supported in the literature by other studies. However, the digestion rate was not out of the range of digestion rates observed with other proteins. For example, soy protein digests slower than whey proteins, but faster than casein. Therefore, we expect that WP-CP particles would still digest faster than soy and casein, which are the two most predominant protein sources other than whey protein used in making bars. If bar formulators intended to use whey protein isolate due to the rapid digestibility, WP-CP particles could still be an option. If one wanted to prolong the effective release of proteins within a bar, WP-CP particles could be exploited in this way, as it could be used as an intermediate digesting rate protein or balanced with another protein in a bar formulation. Ultimately, the amino acid composition, which is what gives whey proteins their high protein quality, would be maintained, and the ingredient would have the added health benefit of the polyphenols. Polyphenols have been shown to impart anti-inflammatory, anti-cancer, and cardiovascular protective properties. In conclusion, the use of whey protein-cranberry polyphenol particles allowed for a reduction in bar hardness. It was hypothesized that protein-polyphenol particles proved a less soluble source of protein, which acted as an inert particle, less likely to interact and cause bar hardening during storage. The digestion rate of the whey protein was decreased in the presence of polyphenols, but remains greater than soy protein and therefore allows for control of protein digestibility. This permits the option to substitute a fraction of WPI with WP-CP particles in a bar formulation to provide both a quick and slow digesting protein, as well as a softer bar. Future work is needed to confirm this proposed mechanism. This project has shown a way to provide additional health benefits and improve the shelf life of protein bars to expand and extend their usefulness in space flight.
Bibliography Type:	Description: (Last Updated: 09/28/2016)
Dissertations and Theses	Schneider ME. "Formation of whey protein-polyphenol aggregates and incorporation into various food structures." Dissertation, North Carolina State University, March 2016. , Mar-2016